

SRTF (Shortest Remaining Time first)

Criteria:- BT

• Tie-breaker: FCFS

• Type:- Preemptive

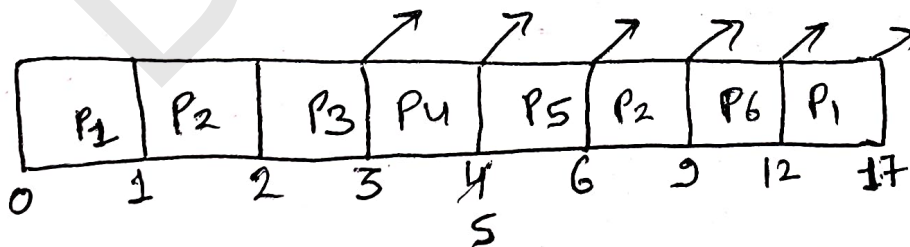
Q:-

Process	Arrival Time	Burst Time
P ₁	0	6
P ₂	1	4
P ₃	2	1
P ₄	3	2
P ₅	4	1
P ₆	5	3

Process	BT
P ₁	6 5
P ₂	4 3
P ₃	1 0
P ₄	2 0
P ₅	1 0
P ₆	3

Pending Queue:-

@ Time 0	P ₁
@ Time 1	P ₁ , P ₂
@ Time 2	P ₁ , P ₂ , P ₃
@ Time 3	P ₁ , P ₂ , P ₄
@ Time 4	P ₁ , P ₂ , P ₄ , P ₅
@ Time 5	P ₁ , P ₂ , P ₆



Process	AT	BT	CT	TAT	WT	RT
P ₁	0	6	17	17	11	0
P ₂	1	4	9	8	4	0
P ₃	2	1	3	1	0	0
P ₄	3	2	5	2	0	0
P ₅	4	1	6	2	1	1
P ₆	5	3	12	7	4	4

$$\text{Avg TAT} = 37/6 \Rightarrow 6.16$$

$$\text{Avg WT} = 20/6 \Rightarrow 3.33$$

$$\text{SL} = 17 - 0 \Rightarrow 17$$

$$T_n = 6/17$$

Response Time = ?

How many context switch?

Q: Consider three CPU intensive processes, which require 10, 20, and 30 time units & arrival time 0, 2, 6 respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? (GATE-2006)

Do not count the context switches at time zero & at the end.

(A) 1

(B) 3

(C) 2

(D) 4

Q: An operating system uses shortest remaining time first scheduling algo for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times & CPU burst time (in ms)
(GATE - 2014)

Process	Arrival time	Burst time
P ₁	0	12
P ₂	2	4
P ₃	3	6
P ₄	8	5

Avg WT (in ms) = ?

⇒ Problems with SJF & SRTF :-

- Starvation
- No fairness
- Practical implementation is not possible